Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology

School of Computing

Department of Computer Science and Engineering

10211CS224 / FULL STACK APPLICATION DEVELOPMENT (JAVA)

PROJECT REVIEW : MILESTONE PROJECT – 1

Date: 26-03-2026

Subscription-Based Membership Engine with Tiered Logic

SDG: SDG No – SDG Title(Example: SDG 4 – Quality Education)

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INTRODUCTION

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- Many digital platforms today use subscription models to provide premium services.
- Managing different membership levels manually is difficult and error-prone.
- Businesses require automation to reduce errors and improve efficiency.
- Subscription systems are widely used in OTT, e-learning, and SaaS platforms.
- This project develops a Subscription-Based Membership Engine with Tiered Logic.
- The system supports Basic, Premium, and Pro plans.
- Each tier provides different features and access levels.
- It manages subscriptions, renewals, upgrades, and expiry.

PROBLEM STATEMENT

- Many organizations still manage memberships manually or using basic systems.
- This leads to errors in billing, user access, and subscription tracking.
- Existing systems do not support flexible tier-based subscription models.
- Monitoring user upgrades, renewals, and expiry becomes difficult.
- Data security and authentication are often not strong.
- Businesses need real-time tracking and automated access control.
- Lack of centralized management reduces efficiency and productivity.
- Therefore, an automated and scalable subscription management system is required.

OBJECTIVE(S)

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- To design and develop an automated Subscription-Based Membership System with tiered access control.

- To implement secure user authentication, subscription management, and payment handling for improved efficiency.

- To design and develop an automated Subscription-Based Membership System with tiered access and secure user authentication.

- To improve system efficiency and evaluate the performance of subscription, billing, and access control mechanisms.

PROPOSED SYSTEM

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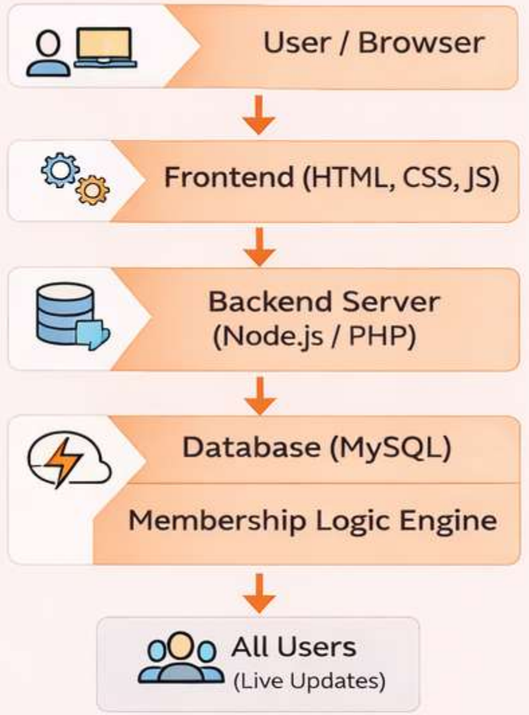
- The system provides an automated solution for managing subscriptions and memberships.
- Users can register, log in, and select suitable subscription plans.
- It supports multiple tiers such as Basic, Premium, and Pro.
- Each tier offers different features and access levels.
- Secure authentication and role-based access control are implemented.
- The system manages subscription upgrades, renewals, and expiry.
- Admin can monitor users, plans, and transaction details.
- Real-time tracking and notifications improve user experience

PROPOSED SYSTEM ARCHITECTURE



User/Browser: Users interact with the system through a web interface.

Frontend: Built using HTML, CSS, and Javascript to handle user inputs and UI.

Backend Server: Processes user requests, handles business logic, and manages subscriptions.

Database (MySQL): Stores user details, subscription plans, and payment information.

Membership Logic Engine: Core component that manages tier-based access, upgrades, and permissions.

Output: Updated membership status and access rights are reflected instantly to users.

WORKFLOW

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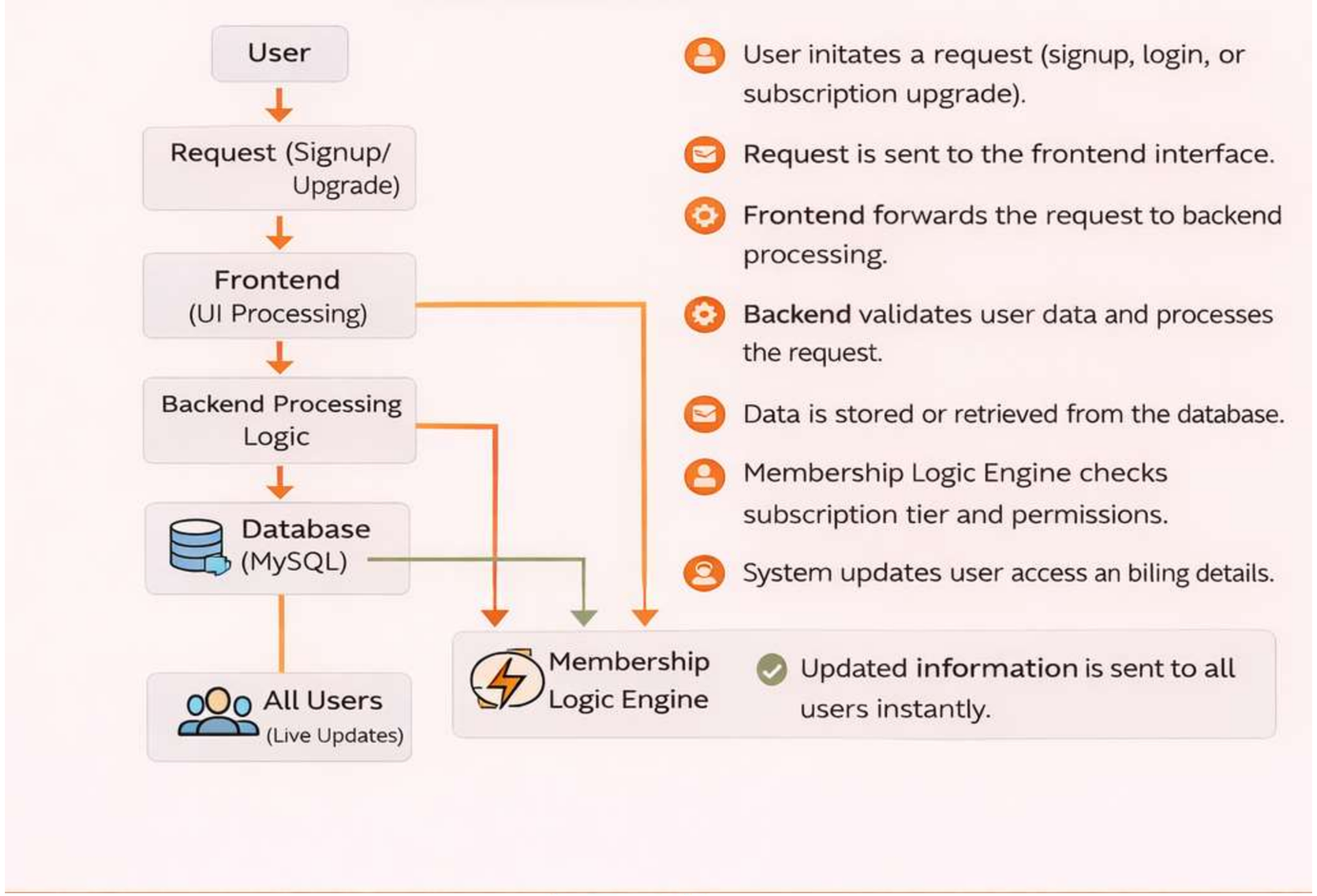

TECHNOLOGIES USED

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Frontend:

- HTML
- CSS
- JavaScript
- Bootstrap

Backend:

- Java
- Spring Boot

Database:

- MySQL

Tools:

- VS Code / Eclipse
- Git & GitHub
- XAMPP:

MODULE DESCRIPTION

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Module 1: User Management
Handles user registration, login, authentication, and profile management with secure access.
Module 2: Core Processing
Allows users to select, upgrade, downgrade, and renew subscription plans based on their needs.
Module 3: Database Management
Processes online payments, maintains transaction records, and manages subscription billing
Module 4: Reporting & Analytics
Provides different features and permissions based on Basic, Premium, and Pro plans.

IMPLEMENTATION SCREENSHOTS

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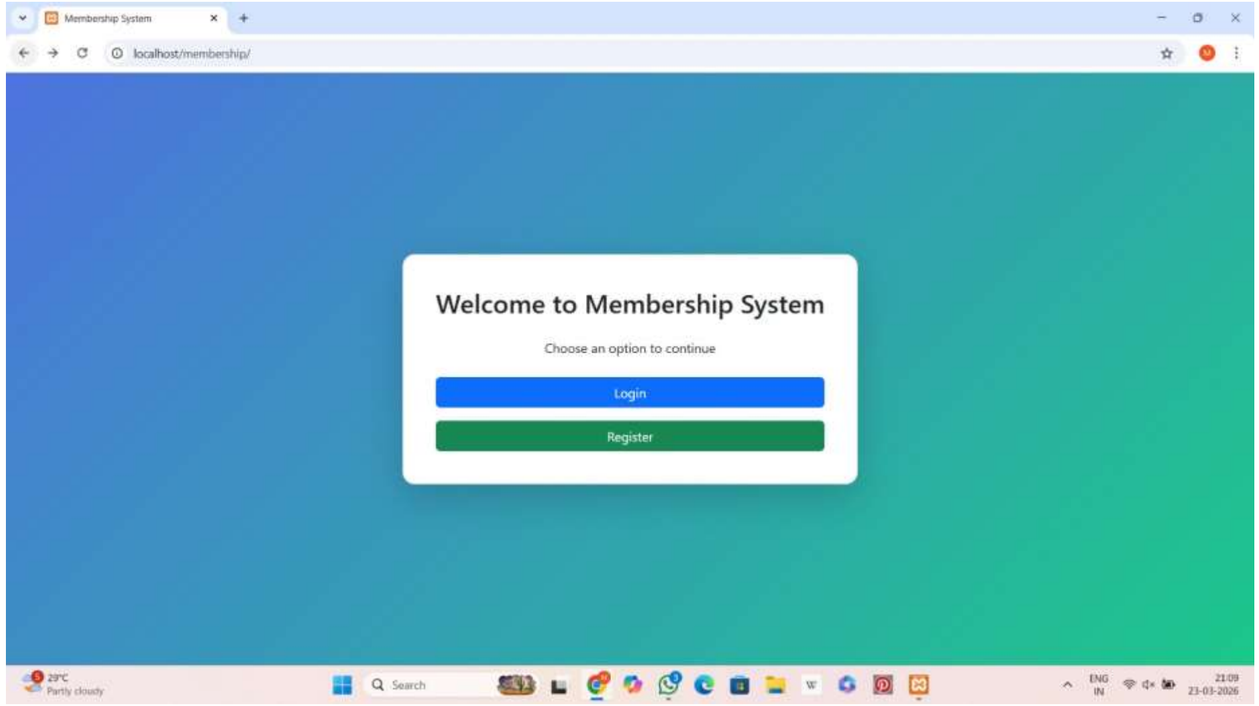
1. Home Page



•This is the landing page of the Membership System with a modern UI and navigation options.
•It allows users to quickly access Login and Register features to start using the system.

IMPLEMENTATION SCREENSHOTS

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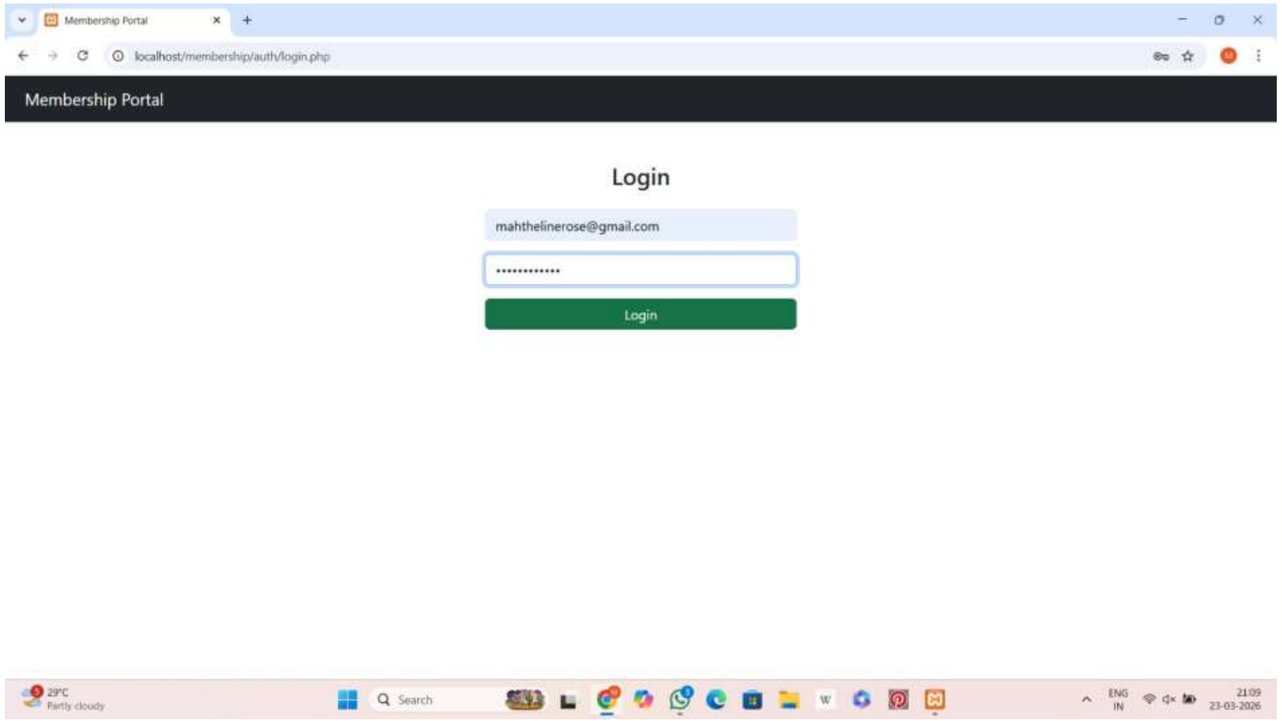
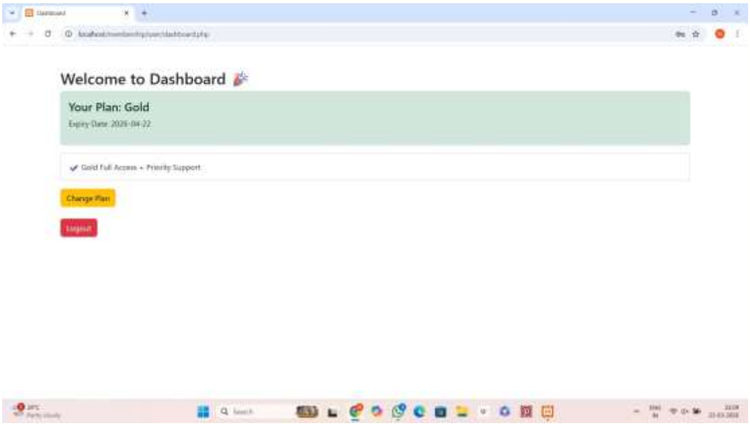
2. Login Page



- This page allows registered users to securely log in using their email and password.
- It validates user credentials and redirects them to the dashboard upon successful login.

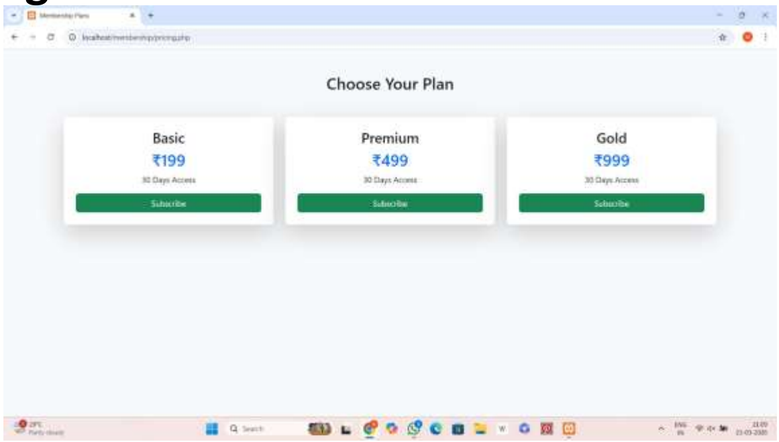
IMPLEMENTATION SCREENSHOTS

3. Dashboard Page



•This page represents the final working system where users can manage their subscription and access features.
•It demonstrates successful implementation of tier-based access and session management.

4. Pricing / Plans Page



•This page displays different subscription plans such as Basic, Premium, and Gold with pricing details.
•Users can choose and subscribe to a plan based on their requirements.

ADVANTAGES & LIMITATIONS

Advantages:

- Reduces manual work and human errors.
- Provides secure user authentication and data protection.
- Supports flexible and scalable subscription plans.
- Enables real-time tracking of subscriptions and payments.
- Improves business efficiency and customer satisfaction.

Limitations:

- Requires a stable internet connection.
- Initial development and setup cost is high.
- System maintenance and security updates are needed.
- Payment gateway integration may be complex.

Thank you